



Predicate Logic

Join the class

Predicate Logic (First Order Logic)

Proposition

P : 2 is greater than 10. (F)

Predicate

$P(x)$: x is greater than 10.
Variable Predicate

$P(2) \Rightarrow (F)$

$P(12) \Rightarrow (T)$

Predicate

When a proposition depends on variable(s), then it becomes a predicate.

Domain

Whatever values the variables can take is generally called as domain.

Predicate

→ Predicate is a proposition except for the fact that it contains variables whose values are to be taken from some universe of discourse.

$C(x): x \text{ is a cricketer.}$
 $E(x): x + 2 \geq 5$

} 1-place predicate

$L(x, y): x \text{ likes } y.$
 $E(x, y): x + y \leq 8$

} 2-place predicates

$f(x, y, z): x + y = z$

$T(p, q, t): p \text{ can teach } q \text{ concept at time } t.$

} 3-place predicates

→ A predicate involving 'n' variables is called
n-place predicate.

$P(x): x > 3, x \text{ is real number.}$

Check for truth $\left\{ \begin{array}{l} P(1) \\ P(2) \\ P(3) \\ P(\pi) \\ P(e) \end{array} \right.$

$P(x): x > 3$, x is real number.

Check for truth

- $P(1) \rightarrow \text{False}$
- $P(2) \rightarrow \text{False}$
- $P(3) \rightarrow \text{False}$
- $P(\pi) \rightarrow \text{True}$
- $P(e) \rightarrow \text{False}$

There are so many real numbers & deciding truth for each is difficult. We need to "quantify" the truth of the predicate $P(x)$.
How much truth is there?

$\exists x P(x)$

$P(x): x^2 \geq 0$, x is real number.

Check for truth

- $P(1) \rightarrow \text{True}$
- $P(2) \rightarrow \text{True}$
- $P(3) \rightarrow \text{True}$
- $P(\pi) \rightarrow \text{True}$
- $P(e) \rightarrow \text{True}$

There are so many real numbers & deciding truth for each is difficult. We need to "quantify" the truth of the predicate $P(x)$.
How much truth is there?

? $P(x)$

$P(x): x^2 \geq 0$, x is real number.

Check for truth

- $P(1) \rightarrow \text{True}$
- $P(2) \rightarrow \text{True}$
- $P(3) \rightarrow \text{True}$
- $P(\pi) \rightarrow \text{True}$
- $P(e) \rightarrow \text{True}$

There are so many real numbers & deciding truth for each is difficult. We need to "quantify" the truth of the predicate $P(x)$.
How much truth is there?

$\forall x P(x)$

$Q(x, y): x + y = 1$ { This is a predicate. }

Is $Q(1, 1)$ true?

$Q(1, 1): 1 + 1 = 1$ { Now, it is a proposition. }

Truth value: False

Is $Q(-2, 1)$ true?

Converting Predicates into Proposition

→ Using Substitution

Quantifiers

→ Universal quantifier

$\forall$

for all

→ Existential quantifier

$\exists$

there exists

or

for some

e.g. $U = \{1, 2\}$

$P(x) : x + 5 \leq 7$

For all x , $P(x)$ is true.

i.e., $\forall x P(x)$

Quantification of truth
→ Always True

e.g. $U = \{1, 2, 3\}$

$P(x) : x + 5 \leq 7$

Quantification { For some x , $P(x)$ is true.
There exists x such that $P(x)$ is true.
There exists atleast one x such that $P(x)$ is true.
i.e., $\exists x P(x)$ $\rightarrow$ Always True

Every student in this class studied Mathematics.

Case I:

U : Set of students in this class.

$M(x)$: x studied Mathematics.

$$\underline{\underline{\forall x M(x)}}$$

For all x , $M(x)$.

Case II:

U : Set of all people.

$S(x)$: x is student in this class.

$M(x)$: x studied Mathematics.

$$\underline{\underline{\forall x [S(x) \rightarrow M(x)]}}$$

For all x ,
if $S(x)$ then $M(x)$.

Negating Quantified Predicates

$$\textcircled{1} \neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\textcircled{2} \neg \exists x P(x) \equiv \forall x \neg P(x)$$

$$\textcircled{3} \overbrace{\forall x \neg P(x)}^{\curvearrowleft} \equiv \neg \exists x P(x)$$

$$\textcircled{4} \overbrace{\exists x \neg P(x)}^{\curvearrowleft} \equiv \neg \forall x P(x)$$

Example 1: Rule (1)

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

Statement:

"All students passed the exam." $\rightarrow \forall x P(x)$, where $P(x)$: " x passed the exam."

Negation:

"Not all students passed the exam." $\rightarrow \neg \forall x P(x)$.

Equivalent form:

"There exists at least one student who did not pass the exam." $\rightarrow \exists x \neg P(x)$.

Example 2: Rule (2)

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Statement:

"There is a student who scored full marks." $\rightarrow \exists x P(x)$, where $P(x)$: " x scored full marks."

Negation:

"There is no student who scored full marks." $\rightarrow \neg \exists x P(x)$.

Equivalent form:

"Every student did not score full marks." $\rightarrow \forall x \neg P(x)$.

Example 3: Rule (3)

$$\forall x \neg P(x) \equiv \neg \exists x P(x)$$

Statement:

"Every student failed the exam." $\rightarrow \forall x \neg P(x)$, where $P(x)$: " x passed the exam."

Equivalent form:

"There does not exist a student who passed the exam." $\rightarrow \neg \exists x P(x)$.

Example 4: Rule (4)

$$\exists x \neg P(x) \equiv \neg \forall x P(x)$$

Statement:

"There exists a student who is not punctual." $\rightarrow \exists x \neg P(x)$, where $P(x)$: " x is punctual."

Equivalent form:

"Not all students are punctual." $\rightarrow \neg \forall x P(x)$.

Aristotle Forms

All P's are Q's.

$$\forall x [P(x) \rightarrow Q(x)]$$

Note: $\forall$ goes with $\rightarrow$

Some P's are Q's.

$$\exists x [P(x) \wedge Q(x)]$$

Note: $\exists$ goes with $\wedge$

Not all P's are Q's.

$$\equiv \neg (\text{All P's are Q's})$$

$$\equiv \neg \{ \forall x [P(x) \rightarrow Q(x)] \}$$

$$\equiv \neg \{ \forall x [\neg P(x) \vee Q(x)] \}$$

$$\equiv \exists x [P(x) \wedge \neg Q(x)]$$

$$\equiv \text{Some P's are not Q's.}$$

No P's are Q's.

$$\equiv \neg (\text{Some P's are Q's})$$

$$\equiv \neg \{ \exists x [P(x) \wedge Q(x)] \}$$

$$\equiv \forall x [\neg P(x) \vee \neg Q(x)]$$

$$\equiv \forall x [P(x) \rightarrow \neg Q(x)]$$

$\equiv$ All P's are not Q's.

Explain why one of these statements is true
and the other is false:

① $(\forall \text{ man } M) (\exists \text{ woman } W) \mid W \text{ is mother of } M$

② $(\exists \text{ woman } W) (\forall \text{ man } M) \mid W \text{ is mother of } M$

Explain why one of these statements is true
and the other is false:

① $(\forall \text{ man } M) (\exists \text{ woman } W) \mid W \text{ is mother of } M$

True

② $(\exists \text{ woman } W) (\forall \text{ man } M) \mid W \text{ is mother of } M$

False

Reason: Order of quantifiers
matter.

① $(\forall n \text{ integer}, n \geq 2) (\exists p \text{ prime}) \quad n/p \text{ is an integer.}$

② $(\exists p \text{ prime}) (\forall n \text{ integer}, n \geq 2) \quad n/p \text{ is an integer.}$

① $(\forall n \text{ integer}, n \geq 2) (\exists p \text{ prime}) \quad n/p \text{ is an integer.}$

True

② $(\exists p \text{ prime}) (\forall n \text{ integer}, n \geq 2) \quad n/p \text{ is an integer.}$

False

Reason: Order of quantifiers
matter.

$$P(x): x^2 < 10$$

Domain of x consist of the positive integers less than 4.

What is the truth value of $\forall x P(x)$?

$$P(x): x^2 < 10$$

Domain of x consist of the positive integers less than 4.

What is the truth value of $\forall x P(x)$?

True

Q. Every student in NST has visited either
Canada or Mexico.

Q. Every student in NST has visited either Canada or Mexico.

Domain of x is all students of NST.

$C(x)$: x has visited Canada.

$M(x)$: x has visited Mexico.

$$\forall x (C(x) \vee M(x))$$

Q. Every student in NST has visited either Canada or Mexico.

Domain of x is all students in the world.

$S(x)$: x is a student at NST

$C(x)$: x has visited Canada.

$M(x)$: x has visited Mexico.

$$\forall x [S(x) \longrightarrow (C(x) \vee M(x))]$$

Some real numbers are rational.

Some real numbers are rational.

$Re(x)$: x is real number.

$Ra(x)$: x is rational number.

$\exists x [Re(x) \wedge Ra(x)]$

Not all that glitters is gold.

Not all that glitters is gold.

$\equiv \neg (\text{All that glitters is gold})$

$\equiv \neg \{ \forall x [Gl(x) \rightarrow Go(x)] \}$

$\equiv \neg \{ \forall x [\neg Gl(x) \vee Go(x)] \}$

$\equiv \exists x [Gl(x) \wedge \neg Go(x)]$

$\equiv$ There exists a metal that glitters and is not gold.

None of my friends are perfect.

None of my friends are perfect.

$\equiv \neg (\text{Some of my friends are perfect})$

$\equiv \neg \{ \exists x [F(x) \wedge P(x)] \}$

$\equiv \forall x [\neg F(x) \vee \neg P(x)]$

$\equiv \forall x [F(x) \rightarrow \neg P(x)]$

$\equiv \text{All of my friends are not perfect.}$

All purple mushrooms are poisonous. [Combined Sense]

All purple mushrooms are poisonous. [Combined Sense]

$P_u(x)$: x is purple.

$M(x)$: x is mushroom.

$P_o(x)$: x is poisonous.

For all x , if x is purple and x is mushroom,
then x is poisonous. $\forall x [(P_u(x) \wedge M(x)) \rightarrow P_o(x)]$

Gold and silver ornaments are precious. [Separate sense]

Gold and silver ornaments are precious. [Separate sense]

$G(x)$: x is gold ornament.

$S(x)$: x is silver ornament.

$P(x)$: x is precious.

For all x , if x is gold or silver ornament, then x is precious.

$$\begin{aligned} & \forall x [(G(x) \rightarrow P(x)) \wedge (S(x) \rightarrow P(x))] \\ & \equiv \forall x [(G(x) \vee S(x)) \rightarrow P(x)] \end{aligned}$$

There is exactly one apple.

There is exactly one apple.

$$\exists x [A(x) \wedge \forall y [A(y) \rightarrow y = x]]$$

There exists x such that x is an apple & there exists y such that if y is an apple, then $y = x$.

There are exactly two apples.

There are exactly two apples.

$$\exists x \exists y [A(x) \wedge A(y) \wedge x \neq y \wedge \\ \forall z [A(z) \rightarrow z = x \vee z = y]]$$

There are at least two apples.

There are at least two apples.

$$\exists x \exists y [A(x) \wedge A(y) \wedge x \neq y]$$

There is atmost one apple.

There is atmost one apple.

Atmost one $\equiv \neg(\text{Atleast two})$

$\neg(\text{There are atleast two apples.})$

There is atmost one apple.

$$\text{Atmost one} \equiv \neg(\text{Atleast two})$$

$$\neg(\text{There are atleast two apples.})$$

$$\equiv \neg \exists x \exists y [A(x) \wedge A(y) \wedge x \neq y]$$

$$\equiv \forall x \forall y [\neg (A(x) \wedge A(y)) \vee x = y]$$

$$\equiv \forall x \forall y [(A(x) \wedge A(y)) \longrightarrow x = y]$$

Provide feedback

95



Thank You!